# Branching of the Hydrogen Cascade after Lyman-Series Absorption

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## 1. Introduction and Assumptions

A Lyman-series absorption event promotes the hydrogen electron from the ground state to an excited $p$ level,

$$1s \longrightarrow np, \qquad n \geq 2, \tag{1}$$

which corresponds to Ly$\alpha$ ($n = 2$), Ly$\beta$ ($n = 3$), Ly$\gamma$ ($n = 4$), and so on. Once in the $np$ level, the electron decays radiatively. The allowed electric-dipole downward channels from $np$ are to the $ms$ states ($m < n$) and to the $md$ states ($3 \leq m < n$). Each downward step lands the electron in a new bound level, from which it decays again; the process repeats until the electron reaches the ground state. We call the complete sequence of radiative transitions from the initially excited $np$ level down to $1s$ the *cascade*.

Two terminal channels close the cascade and are physically distinct:

(a) **Direct Lyman emission.** Whenever the electron reaches an $n'p$ level, it may decay directly to the ground state, $n'p \rightarrow 1s$, emitting a Lyman-series photon. This ends the cascade on the ground state.

(b) **Two-photon continuum.** The cascade may instead route the electron into the metastable $2s$ state. Because $2s \rightarrow 1s$ is electric-dipole forbidden, this level decays by *two-photon* emission, $A(2s \rightarrow 1s) = 8.229\ \mathrm{s}^{-1}$, producing a broad continuum rather than a spectral line.

Throughout we adopt the low-density nebular (Case-B-like) regime: a pure radiative cascade with no collisional $l$-redistribution, and with the $2s$ state always decaying via two-photon emission. In this limit the branching of the cascade is fixed entirely by the spontaneous transition probabilities of hydrogen, independent of density. A useful example is Ly$\beta$: absorption into $3p$ can decay directly back to $1s$ (re-emitting Ly$\beta$) or route through $3p \rightarrow 2s$, which is a component

of H$\alpha$. In the latter case the Ly$\beta$ absorption is physically converted into an H$\alpha$ photon ($3p \to 2s$) plus a two-photon ($2s \to 1s$) continuum.

## 2. Lyman-Series Oscillator Strengths

Table 1 lists the absorption oscillator strengths $f(1s \to np)$ of the hydrogen Lyman series, together with the transition wavelengths and the ratios relative to Ly$\alpha$. For large $n$ the oscillator strength falls off asymptotically as $n^{-3}$, so successive Lyman lines weaken rapidly.

It is important to note that the line-center opacity scales as the product $f\,\lambda$, not as $f$ alone. Because the higher Lyman lines lie at shorter wavelengths, the opacity ratio is slightly smaller than the oscillator-strength ratio. For Ly$\beta$ relative to Ly$\alpha$,

$$\frac{\tau_0(\mathrm{Ly}\beta)}{\tau_0(\mathrm{Ly}\alpha)} = 0.1901 \times \frac{1025.72}{1215.67} = 0.160, \tag{2}$$

rather than 0.190.

Table 1: Lyman-series absorption oscillator strengths $f(1s \to np)$.

| $n$ | line | $\lambda$ [Å] | $f$ | $f/f(\mathrm{Ly}\alpha)$ |
|---|---|---|---|---|
| 2 | Ly$\alpha$ | 1215.67 | 0.4162 | 1.00000 |
| 3 | Ly$\beta$ | 1025.72 | 0.07910 | 0.19006 |
| 4 | Ly$\gamma$ | 972.54 | 0.02899 | 0.06966 |
| 5 | Ly$\delta$ | 949.74 | 0.01394 | 0.03349 |
| 6 | Ly$\epsilon$ | 937.80 | 0.007799 | 0.01874 |
| 7 | $1 \to 7$ | 930.75 | 0.004814 | 0.01157 |
| 8 | $1 \to 8$ | 926.23 | 0.003183 | 0.00765 |
| 9 | $1 \to 9$ | 923.15 | 0.002216 | 0.00532 |
| 10 | $1 \to 10$ | 920.96 | 0.001605 | 0.00386 |

## 3. Method and Validation

The spontaneous transition probabilities $A(nl \to n'l')$ were computed from the exact hydrogenic dipole radial integrals and calibrated on the analytically known value $A(2p \to 1s) = 6.2649 \times 10^8$ s$^{-1}$, which fixes the Einstein-$A$ prefactor. The absorption oscillator strengths follow from

$$f(1s \to np) = \tfrac{2}{3}\,\Delta E\,\left|\langle np|\,r\,|1s\rangle\right|^2, \tag{3}$$

evaluated in atomic units, where $\Delta E$ is the transition energy. Every computed value reproduces the literature to four or five significant figures. Table 2 demonstrates this agreement for a representative set of transitions spanning the Lyman, Balmer, and Paschen series; the computed and literature $A$ coefficients agree to within 0.0–0.4%.

Table 2: Validation of computed Einstein $A$ coefficients ($s^{-1}$).

| transition | series | computed | literature |
|---|---|---|---|
| $A(2p \to 1s)$ | Ly$\alpha$ | $6.2649 \times 10^8$ | $6.2649 \times 10^8$ |
| $A(3p \to 1s)$ | Ly$\beta$ | $1.6725 \times 10^8$ | $1.6725 \times 10^8$ |
| $A(3s \to 2p)$ | H$\alpha$ | $6.3136 \times 10^6$ | $6.3132 \times 10^6$ |
| $A(3p \to 2s)$ | H$\alpha$ | $2.2448 \times 10^7$ | $2.2449 \times 10^7$ |
| $A(3d \to 2p)$ | H$\alpha$ | $6.4651 \times 10^7$ | $6.4651 \times 10^7$ |
| $A(4p \to 1s)$ | Ly$\gamma$ | $6.8187 \times 10^7$ | $6.8186 \times 10^7$ |
| $A(4d \to 2p)$ | H$\beta$ | $2.0625 \times 10^7$ | $2.0625 \times 10^7$ |
| $A(4f \to 3d)$ | Pa$\alpha$ | $1.3788 \times 10^7$ | $1.3785 \times 10^7$ |

## 4. Direct Branching Ratios of the $np$ Levels

Table 3 gives the direct branching ratios $b(np \to n'l') = A(np \to n'l')/A_{\mathrm{tot}}(np)$ of the initially excited $np$ level, expressed as percentages. Here $A_{\mathrm{tot}}$ is the total spontaneous decay rate and $\tau = 1/A_{\mathrm{tot}}$ the radiative lifetime. These are the single-step probabilities out of the $np$ level, not yet the full cascade outcome (Section 5.).

Every $np$ level has exactly three kinds of downward channel, and the table groups them so that the three add up to 100 %: (i) the resonance line $np \to 1s$, a Lyman photon straight back to the ground state; (ii) $np \to 2s$, which populates the metastable $2s$ level and therefore feeds the two-photon continuum; (iii) the subordinate decays to any lower level with $m \geq 3$ (the *ms* and *md* states $3s, 3d, 4s, 4d, \ldots$), from which the cascade continues. The last two columns single out the two largest subordinate channels, $np \to 3s$ and $np \to 3d$; these are *already included* in the $m \geq 3$ column and are listed only to show where the subordinate weight goes.

From $3p$ only the first two channels exist, because no *md* state with $m \geq 3$ lies below $n = 3$. Subordinate decay first appears at $4p$ and grows with $n$: the Lyman branch $np \to 1s$ falls monotonically from 100 % at $2p$ to 78.5 % at $10p$, the subordinate fraction rises to about 10 %, and the two-photon feed $np \to 2s$ stays nearly constant near 11.5–11.9 % across the whole series.

## 5. Cascade Outcome Probabilities

Following the cascade all the way to the ground state, every absorbed Lyman-series photon ends in exactly one of the two terminal channels of Section 1: either the $2s$ two-photon continuum, or a direct Lyman photon $n'p \to 1s$. Table 4 gives these total probabilities as a function of the initially excited level.

The most striking feature is the near-constancy of the outcome. The probability of ending in the two-photon continuum is

$$P(\text{two-photon}) \approx 0.118 \quad (11.8\%), \tag{4}$$

essentially independent of the initial principal quantum number $n$, and correspondingly

$$P(\text{direct Lyman}) \approx 0.88. \tag{5}$$

Table 3: Direct (single-step) decay of the $np$ levels, as branching percentages. $\tau = 1/A_{\rm tot}$ is the radiative lifetime. The three columns under "Branching ratio" are the probabilities that the first emitted photon is the Lyman line $np \to 1s$, feeds the metastable $2s$ (and hence the two-photon continuum), or goes to a subordinate level with lower principal quantum number $m \geq 3$; *these three sum to* 100 %. The two right-hand columns break out the two largest subordinate channels, $np \to 3s$ and $np \to 3d$, which are already counted in the $m \geq 3$ column. "–" marks a channel that does not exist.

| | | | Branching ratio (%) | | | Largest subord. (%) | |
|---|---|---|---|---|---|---|---|
| $np$ | $\tau$ [ns] | $A_{\rm tot}$ [s$^{-1}$] | $\to 1s$ (Lyman) | $\to 2s$ (2-photon) | $\to m \geq 3$ (subord.) | $\to 3s$ | $\to 3d$ |
| $2p$ | 1.596 | $6.2649 \times 10^8$ | 100.00 | – | – | – | – |
| $3p$ | 5.271 | $1.8970 \times 10^8$ | 88.17 | 11.83 | – | – | – |
| $4p$ | 12.305 | $8.1267 \times 10^7$ | 83.90 | 11.90 | 4.20 | 3.77 | 0.43 |
| $5p$ | 23.789 | $4.2037 \times 10^7$ | 81.78 | 11.77 | 6.45 | 3.90 | 0.36 |
| $6p$ | 40.819 | $2.4499 \times 10^7$ | 80.53 | 11.67 | 7.80 | 3.90 | 0.32 |
| $7p$ | 64.492 | $1.5506 \times 10^7$ | 79.72 | 11.59 | 8.69 | 3.89 | 0.30 |
| $8p$ | 95.909 | $1.0427 \times 10^7$ | 79.17 | 11.53 | 9.30 | 3.87 | 0.29 |
| $9p$ | 136.169 | $7.3438 \times 10^6$ | 78.77 | 11.49 | 9.74 | 3.86 | 0.28 |
| $10p$ | 186.372 | $5.3656 \times 10^6$ | 78.47 | 11.46 | 10.07 | 3.85 | 0.27 |

Ly$\alpha$ ($2p$) is the sole exception: $2p$ can only decay to $1s$, so it never routes through $2s$ and always re-emits a Lyman photon. For all higher Lyman lines, roughly one absorbed photon in eight is converted into the two-photon continuum, with the remainder re-emerging as a direct Lyman photon. The two-photon probability actually peaks at $4p$ (0.11897) and then stays remarkably flat, remaining within 0.1180–0.1190 across the entire series through $10p$; the direct-Lyman probability correspondingly sits at $\approx$ 0.881–0.882.

Table 4: Cascade outcome after $1s \to np$ absorption: total probability of the $2s$ two-photon continuum versus a direct Lyman photon.

| initial level | $P$(2-photon via $2s$) | $P$(direct Lyman $n'p \to 1s$) |
|---|---|---|
| $2p$ (Ly$\alpha$) | 0.00000 | 1.00000 |
| $3p$ (Ly$\beta$) | 0.11834 | 0.88166 |
| $4p$ (Ly$\gamma$) | 0.11897 | 0.88103 |
| $5p$ (Ly$\delta$) | 0.11871 | 0.88129 |
| $6p$ (Ly$\epsilon$) | 0.11848 | 0.88152 |
| $7p$ ($1 \to 7$) | 0.11831 | 0.88169 |
| $8p$ ($1 \to 8$) | 0.11818 | 0.88182 |
| $9p$ ($1 \to 9$) | 0.11808 | 0.88192 |
| $10p$ ($1 \to 10$) | 0.11800 | 0.88200 |

## 6. Cascade Route Decomposition

Rather than counting how often each individual transition is emitted, it is far more illuminating to follow the complete *routes* the electron takes from the initially excited $np$ level down to the ground state. A route is a chain $np \to \cdots \to 1s$; the transitions along it are exactly the photons emitted *together* in one de-excitation, and the route probability is the product of the branching ratios of its steps. Because every absorbed photon follows one and only one route to the ground state, the route probabilities sum to unity.

The chains are forced by the dipole selection rule $\Delta l = \pm 1$: only $p$ states connect to the $1s$ ground state, so an electron that reaches an $s$ or $d$ state must first step to a lower $p$ state before it can reach $1s$. This is why $3s$ does not decay to $1s$ directly but must follow $3s \to 2p \to 1s$, and it is what makes the routes fall into a few systematic families.

Table 5 lists the dominant routes, organized into five families: (A) the direct resonance decay $np \to 1s$; (B) the metastable route $np \to 2s \to 1s$, which ends in the two-photon continuum; (C) routes through one subordinate $ns$ state that end on Ly$\alpha$ ($\cdots \to 2p \to 1s$); (D) routes through one subordinate $nd$ state that end on Ly$\alpha$; and (E) routes through an $ns$ state that end on Ly$\beta$ ($\cdots \to 3p \to 1s$). The intermediate steps make the co-emitted photons explicit: e.g. the route $np \to 3s \to 2p \to 1s$ emits, in succession, a subordinate photon, an H$\alpha$ photon ($3s \to 2p$), and finally a Ly$\alpha$ photon ($2p \to 1s$). All routes not listed are collected in the "other routes" residual; each has a probability below $\sim 0.3\%$, and they proliferate with $n$ (the $10p$ cascade has 1157 distinct routes in all).

Reading off the $10p$ column, for example: the absorbed Ly-10 photon is simply re-emitted as $10p \to 1s$ with probability 78.5%; it is converted into the two-photon continuum via $10p \to 2s \to 1s$ with probability 11.5%; and it degrades into an H$\alpha$-then-Ly$\alpha$ pair via $10p \to 3s \to 2p \to 1s$ with probability 3.85%. The route probabilities are remarkably insensitive to the initial level — each family contributes nearly the same fraction whether the absorbed line is Ly$\gamma$ or Ly-10.

## 7. Discussion

The near-constant 11.8% conversion probability has a direct physical interpretation. For Ly$\beta$, an absorption that does not simply re-emit Ly$\beta$ is instead converted, with probability 0.11834, into an H$\alpha$ photon ($3p \to 2s$) plus a $2s \to 1s$ two-photon continuum. The higher Lyman lines behave almost identically: about one absorbed photon in eight is permanently removed from the Lyman series and redistributed into subordinate lines and the two-photon continuum, while the remaining $\sim 88\%$ re-emerges as a direct Lyman photon.

Extending the analysis across the whole Lyman series through Ly-10 ($1 \to 10$) reinforces this picture. The conversion probability to the $2s$ two-photon continuum stays near 11.8–11.9%, essentially independent of which Lyman line is absorbed. What does change with $n$ is the fraction of the absorbed photon's energy that is degraded into subordinate (Balmer, Paschen, ...) lines during the longer cascade: this grows steadily with the initial principal quantum number, so that the total number of emitted photons per absorbed Lyman-series photon rises from exactly 1 at Ly$\alpha$ to $\approx 1.20$ at Ly-10. A higher Lyman-line absorption therefore returns a direct Lyman photon with nearly the same probability regardless of $n$, but it distributes progressively more of the cascade energy into the lower series as $n$ increases.

This branching is central to the treatment of nebular and intergalactic Lyman pumping, and to Ly$\alpha$ radiative transfer more generally. Higher Lyman-series photons that are absorbed are not conservatively scattered: a fixed fraction is

Table 5: Dominant cascade routes after $1s \to np$ absorption. Each row is a route — a chain $np \to \cdots \to 1s$ whose transitions are the photons emitted *together* in one de-excitation — and the entries are the route probability (in %) for each initially excited level. Routes are grouped into families (A)–(E) by their structure; "$1s\,(2\gamma)$" marks the $2s \to 1s$ two-photon continuum. "–" means the route is not open at that $n$. Every column, including the "other routes" residual, sums to 100%.

| route | $3p$ | $4p$ | $5p$ | $6p$ | $7p$ | $8p$ | $9p$ | $10p$ |
|---|---|---|---|---|---|---|---|---|
| *(A) Direct resonance decay (final photon: the Ly-n line)* | | | | | | | | |
| $np \to 1s$ | 88.17 | 83.90 | 81.78 | 80.53 | 79.72 | 79.17 | 78.77 | 78.47 |
| *(B) Via metastable* $2s$ *(two-photon continuum)* | | | | | | | | |
| $np \to 2s \to 1s\,(2\gamma)$ | 11.83 | 11.90 | 11.77 | 11.67 | 11.59 | 11.53 | 11.49 | 11.46 |
| *(C) Via ns, ending on Ly*$\alpha$ ($\to 2p \to 1s$) | | | | | | | | |
| $np \to 3s \to 2p \to 1s$ | – | 3.77 | 3.90 | 3.90 | 3.89 | 3.87 | 3.86 | 3.85 |
| $np \to 4s \to 2p \to 1s$ | – | – | 1.02 | 1.06 | 1.07 | 1.06 | 1.06 | 1.06 |
| $np \to 5s \to 2p \to 1s$ | – | – | – | 0.45 | 0.47 | 0.47 | 0.46 | 0.46 |
| $np \to 6s \to 2p \to 1s$ | – | – | – | – | 0.25 | 0.26 | 0.26 | 0.25 |
| *(D) Via nd, ending on Ly*$\alpha$ ($\to 2p \to 1s$) | | | | | | | | |
| $np \to 3d \to 2p \to 1s$ | – | 0.43 | 0.36 | 0.32 | 0.30 | 0.29 | 0.28 | 0.27 |
| $np \to 4d \to 2p \to 1s$ | – | – | 0.33 | 0.29 | 0.26 | 0.24 | 0.23 | 0.23 |
| $np \to 5d \to 2p \to 1s$ | – | – | – | 0.26 | 0.22 | 0.20 | 0.19 | 0.18 |
| *(E) Via ns, ending on Ly*$\beta$ ($\to 3p \to 1s$) | | | | | | | | |
| $np \to 4s \to 3p \to 1s$ | – | – | 0.64 | 0.67 | 0.67 | 0.67 | 0.67 | 0.66 |
| $np \to 5s \to 3p \to 1s$ | – | – | – | 0.28 | 0.29 | 0.29 | 0.29 | 0.29 |
| *Other routes (each below* 0.3%*)* | | | | | | | | |
| other routes | 0.00 | 0.00 | 0.20 | 0.58 | 1.28 | 1.95 | 2.44 | 2.81 |

destroyed as line photons and converted into the two-photon continuum and lower-series lines (for example, feeding H$\alpha$ and the two-photon continuum from Ly$\beta$). In a Monte Carlo transfer calculation this means that the excitation of $np$ levels by higher Lyman lines cannot be modeled as pure resonance scattering; the cascade branching of Tables 4 and 5 determines how much Lyman flux is conserved and how much is converted, essentially independently of the initial level for $n \geq 3$.